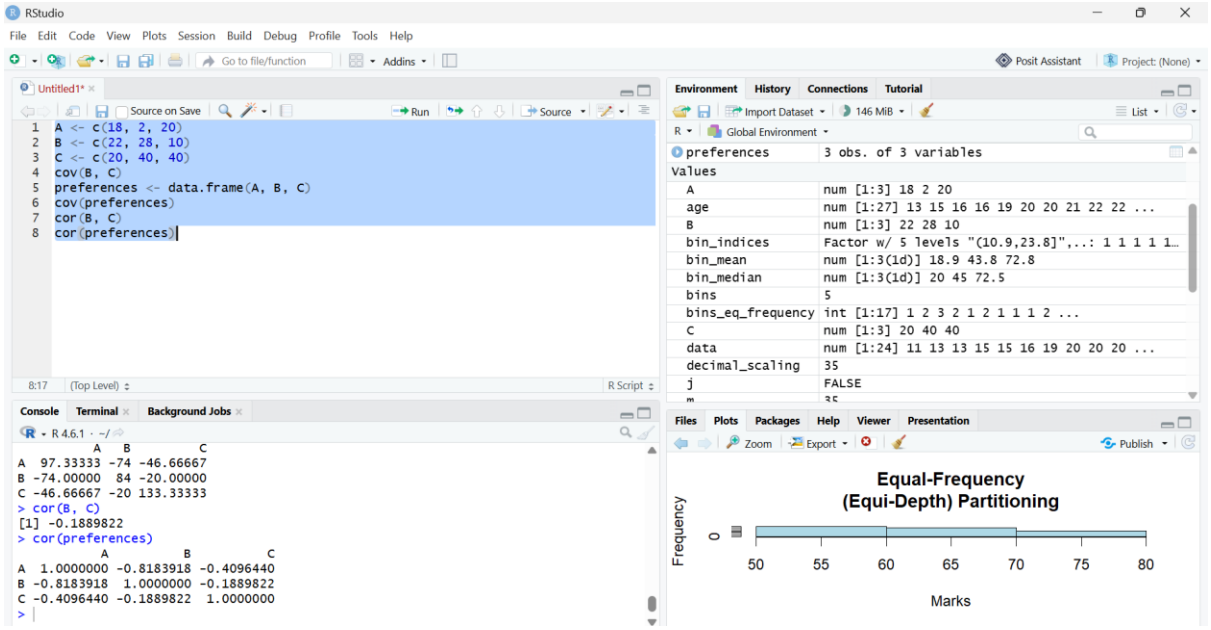
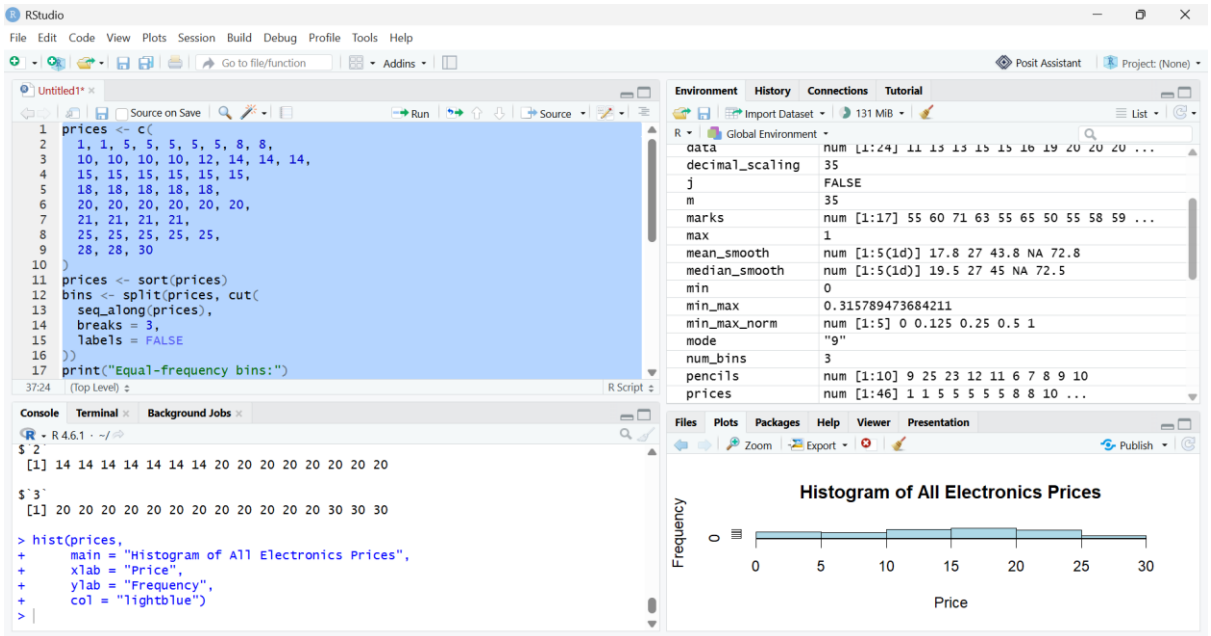
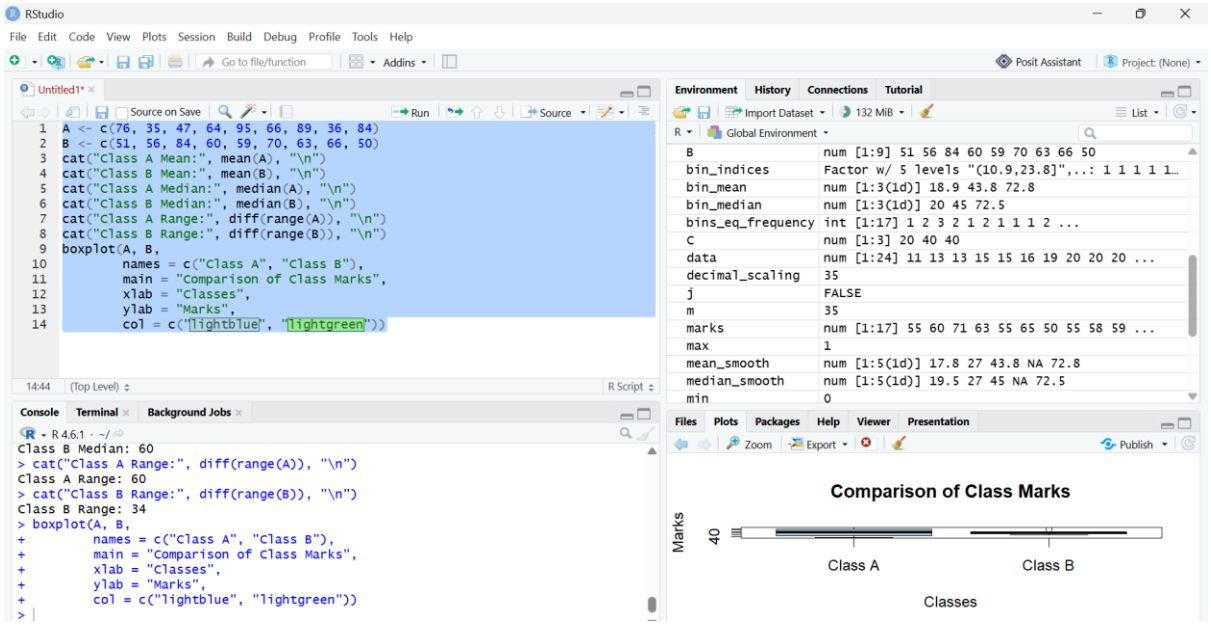
Q1.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
1 A <- c(18, 2, 20)
2 B <- c(22, 28, 10)
3 C <- c(20, 40, 40)
4 cov(B, C)
5 preferences <- data.frame(A, B, C)
6 cov(preferences)
7 cor(B, C)
8 cor(preferences)
Environment History Connections Tutorial
preferences 3 obs. of 3 variables
A num [1:3] 18 2 20
age num [1:27] 13 15 16 16 19 20 20 21 22 22 ...
B num [1:3] 22 28 10
bin_indices Factor w/ 5 levels "(10.9,23.8]",..: 1 1 1 1 1...
bin_mean num [1:3(1d)] 18.9 43.8 72.8
bin_median num [1:3(1d)] 20 45 72.5
bins 5
bins_eq_frequency int [1:17] 1 2 3 2 1 2 1 1 1 2 ...
C num [1:3] 20 40 40
data num [1:24] 11 13 13 15 15 16 19 20 20 20 ...
decimal_scaling 35
j FALSE
Console Terminal Background Jobs
A B C
A 97.33333 -74 -46.66667
B -74.00000 84 -20.00000
C -46.66667 -20 133.33333
> cor(B, C)
[1] -0.1889822
> cor(preferences)
A B C
A 1.0000000 -0.8183918 -0.4096440
B -0.8183918 1.0000000 -0.1889822
C -0.4096440 -0.1889822 1.0000000
Files Plots Packages Help Viewer Presentation
Equal-Frequency
(Equi-Depth) Partitioning
Frequency
0
50 55 60 65 70 75 80
Marks

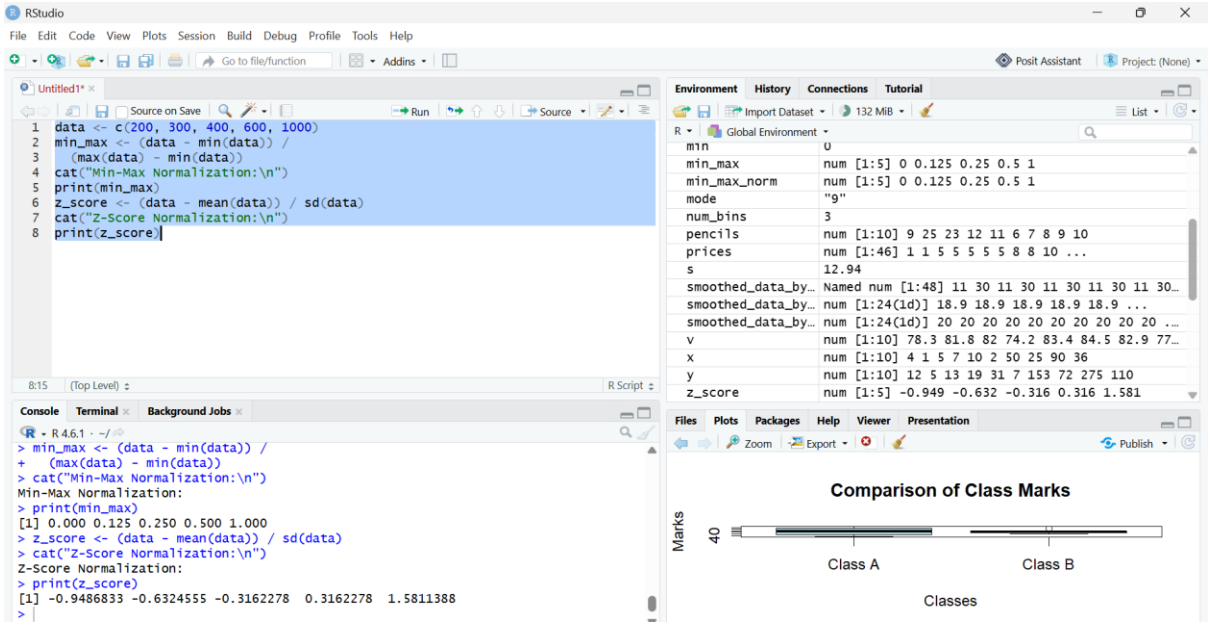
Q2.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
1 prices <- c(
2 1, 1, 5, 5, 5, 5, 8, 8,
3 10, 10, 10, 10, 12, 14, 14, 14,
4 15, 15, 15, 15, 15, 15,
5 18, 18, 18, 18, 18,
6 20, 20, 20, 20, 20, 20,
7 21, 21, 21, 21,
8 25, 25, 25, 25, 25,
9 28, 28, 30
10)
11 prices <- sort(prices)
12 bins <- split(prices, cut(
13 seq_along(prices),
14 breaks = 3,
15 labels = FALSE
16))
17 print("Equal-frequency bins:")
Environment History Connections Tutorial
decimal_scaling 35
j FALSE
m 35
marks num [1:17] 55 60 71 63 55 65 50 55 58 59 ...
max 1
mean_smooth num [1:5(1d)] 17.8 27 43.8 NA 72.8
median_smooth num [1:5(1d)] 19.5 27 45 NA 72.5
min 0
min_max 0.315789473684211
min_max_norm num [1:5] 0 0.125 0.25 0.5 1
mode "9"
num_bins 3
pencils num [1:10] 9 25 23 12 11 6 7 8 9 10
prices num [1:46] 1 1 5 5 5 5 5 8 8 10 ...
Console Terminal Background Jobs
$`2`
[1] 14 14 14 14 14 14 20 20 20 20 20 20 20 20
$`3`
[1] 20 20 20 20 20 20 20 20 20 20 20 20 30 30 30
> hist(prices,
+ main = "Histogram of All Electronics Prices",
+ xlab = "Price",
+ ylab = "Frequency",
+ col = "lightblue")
Files Plots Packages Help Viewer Presentation
Histogram of All Electronics Prices
Frequency
0
0 5 10 15 20 25 30
Price

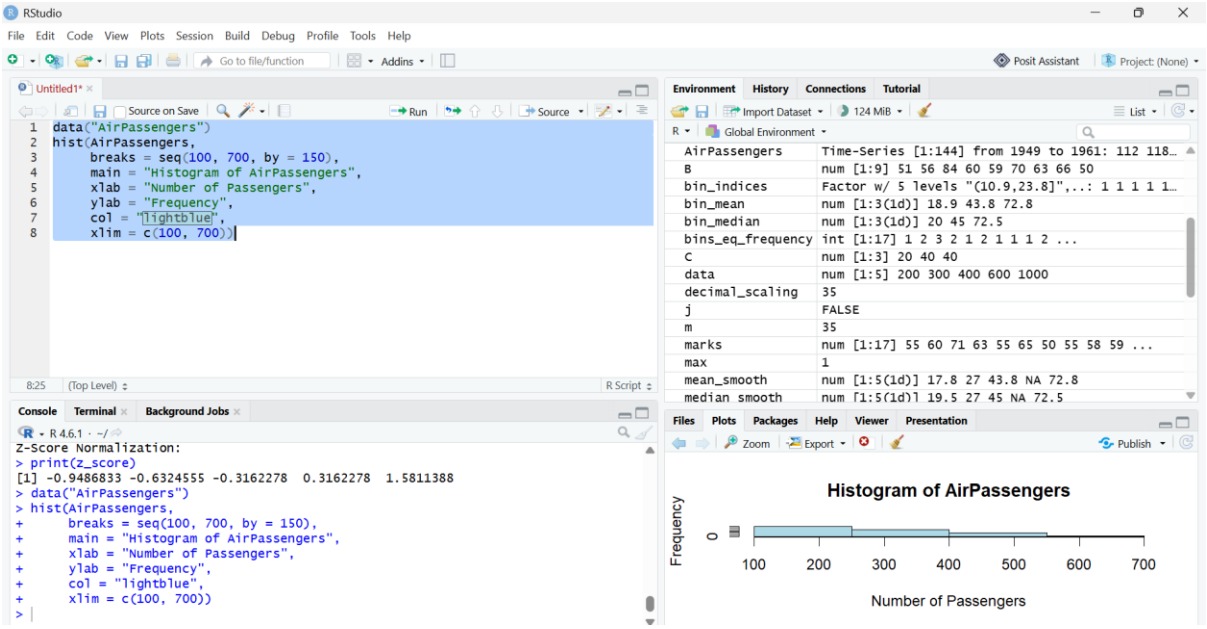
Q3.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
A <- c(76, 35, 47, 64, 95, 66, 89, 36, 84)
B <- c(51, 56, 84, 60, 59, 70, 63, 66, 50)
cat("Class A Mean:", mean(A), "\n")
cat("Class B Mean:", mean(B), "\n")
cat("Class A Median:", median(A), "\n")
cat("Class B Median:", median(B), "\n")
cat("Class A Range:", diff(range(A)), "\n")
cat("Class B Range:", diff(range(B)), "\n")
boxplot(A, B,
names = c("Class A", "Class B"),
main = "Comparison of Class Marks",
xlab = "Classes",
ylab = "Marks",
col = c("lightblue", "lightgreen"))
Class B Median: 60
> cat("Class A Range:", diff(range(A)), "\n")
Class A Range: 60
> cat("Class B Range:", diff(range(B)), "\n")
Class B Range: 34
> boxplot(A, B,
+ names = c("Class A", "Class B"),
+ main = "Comparison of Class Marks",
+ xlab = "Classes",
+ ylab = "Marks",
+ col = c("lightblue", "lightgreen"))
Comparison of Class Marks
Marks
40
Class A
Class B
Classes

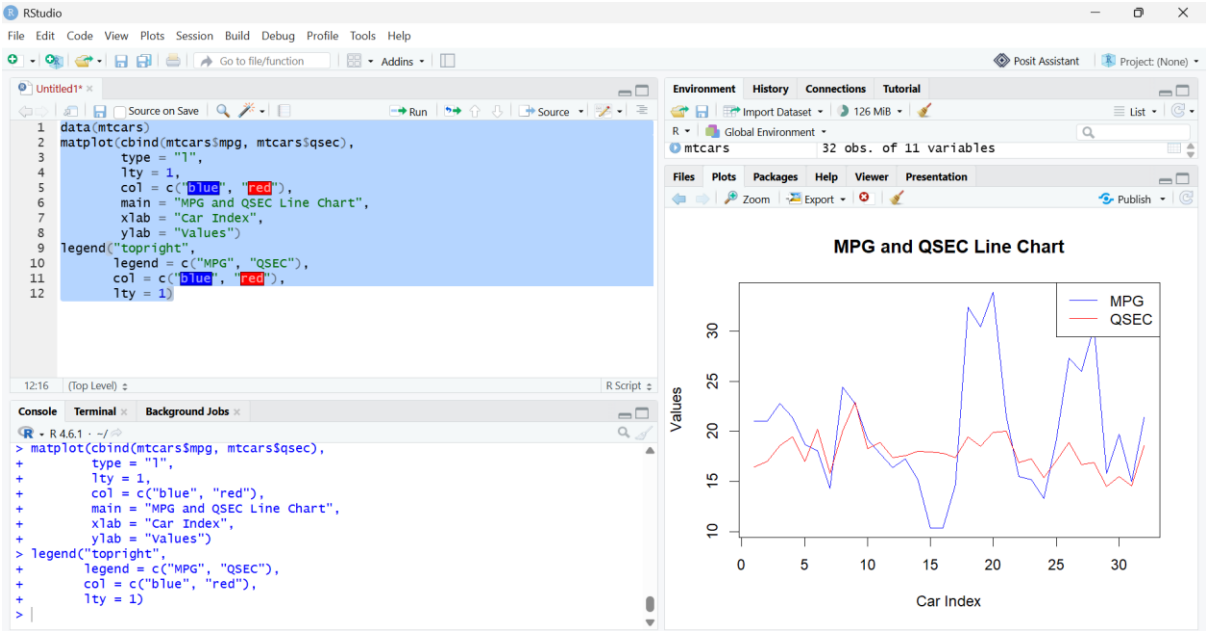
Q4.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
data <- c(200, 300, 400, 600, 1000)
min_max <- (data - min(data)) /
(max(data) - min(data))
cat("Min-Max Normalization:\n")
print(min_max)
z_score <- (data - mean(data)) / sd(data)
cat("Z-Score Normalization:\n")
print(z_score)
> min_max <- (data - min(data)) /
+ (max(data) - min(data))
> cat("Min-Max Normalization:\n")
Min-Max Normalization:
> print(min_max)
[1] 0.000 0.125 0.250 0.500 1.000
> z_score <- (data - mean(data)) / sd(data)
> cat("Z-Score Normalization:\n")
Z-Score Normalization:
> print(z_score)
[1] -0.9486833 -0.6324555 -0.3162278 0.3162278 1.5811388
Comparison of Class Marks
Marks
40
Class A
Class B
Classes

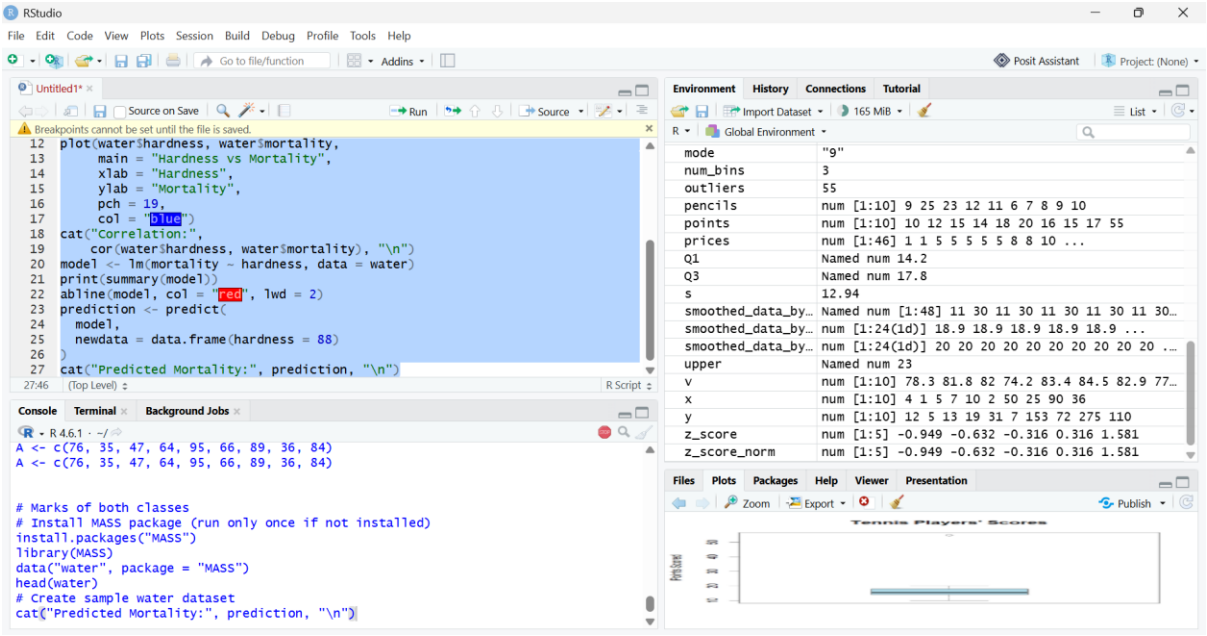
Q5.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
1 data("AirPassengers")
2 hist(AirPassengers,
3 breaks = seq(100, 700, by = 150),
4 main = "Histogram of AirPassengers",
5 xlab = "Number of Passengers",
6 ylab = "Frequency",
7 col = "lightblue",
8 xlim = c(100, 700))
Environment History Connections Tutorial
AirPassengers Time-Series [1:144] from 1949 to 1961: 112 118…
B num [1:9] 51 56 84 60 59 70 63 66 50
bin_indices Factor w/ 5 levels "(10.9,23.8]",..: 1 1 1 1 1…
bin_mean num [1:3(1d)] 18.9 43.8 72.8
bin_median num [1:3(1d)] 20 45 72.5
bins_eq_frequency int [1:17] 1 2 3 2 1 2 1 1 1 2 ...
c num [1:3] 20 40 40
data num [1:5] 200 300 400 600 1000
decimal_scaling 35
j FALSE
m 35
marks num [1:17] 55 60 71 63 55 65 50 55 58 59 ...
max 1
mean_smooth num [1:5(1d)] 17.8 27 43.8 NA 72.8
median_smooth num [1:5(1d)] 19.5 27 45 NA 72.5
Console Terminal Background Jobs
Z-Score Normalization:
> print(z_score)
[1] -0.9486833 -0.6324555 -0.3162278 0.3162278 1.5811388
> data("AirPassengers")
> hist(AirPassengers,
+ breaks = seq(100, 700, by = 150),
+ main = "Histogram of AirPassengers",
+ xlab = "Number of Passengers",
+ ylab = "Frequency",
+ col = "lightblue",
+ xlim = c(100, 700))
Files Plots Packages Help Viewer Presentation
Histogram of AirPassengers
Frequency
Number of Passengers

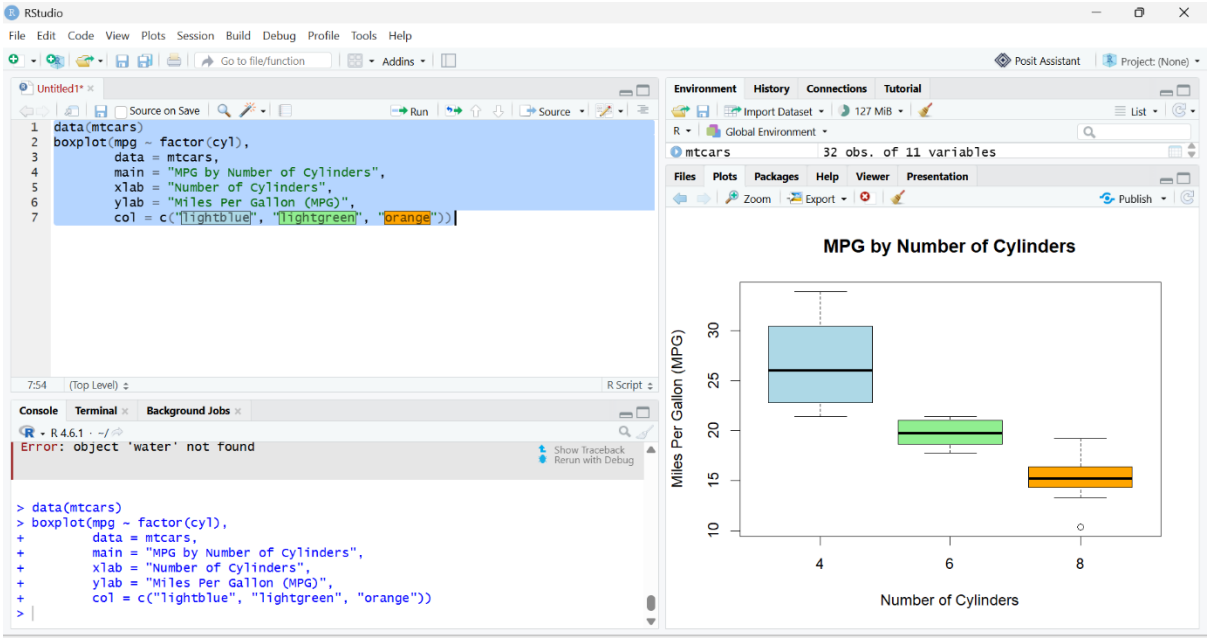
Q6.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
1 data(mtcars)
2 matplot(cbind(mtcars$mpg, mtcars$qsec),
3 type = "l",
4 lty = 1,
5 col = c("blue", "red"),
6 main = "MPG and QSEC Line Chart",
7 xlab = "Car Index",
8 ylab = "Values")
9 legend("topright",
10 legend = c("MPG", "QSEC"),
11 col = c("blue", "red"),
12 lty = 1)
Environment History Connections Tutorial
mtcars 32 obs. of 11 variables
Console Terminal Background Jobs
> matplot(cbind(mtcars$mpg, mtcars$qsec),
+ type = "l",
+ lty = 1,
+ col = c("blue", "red"),
+ main = "MPG and QSEC Line Chart",
+ xlab = "Car Index",
+ ylab = "Values")
> legend("topright",
+ legend = c("MPG", "QSEC"),
+ col = c("blue", "red"),
+ lty = 1)
Files Plots Packages Help Viewer Presentation
MPG and QSEC Line Chart
MPG
QSEC
Values
Car Index

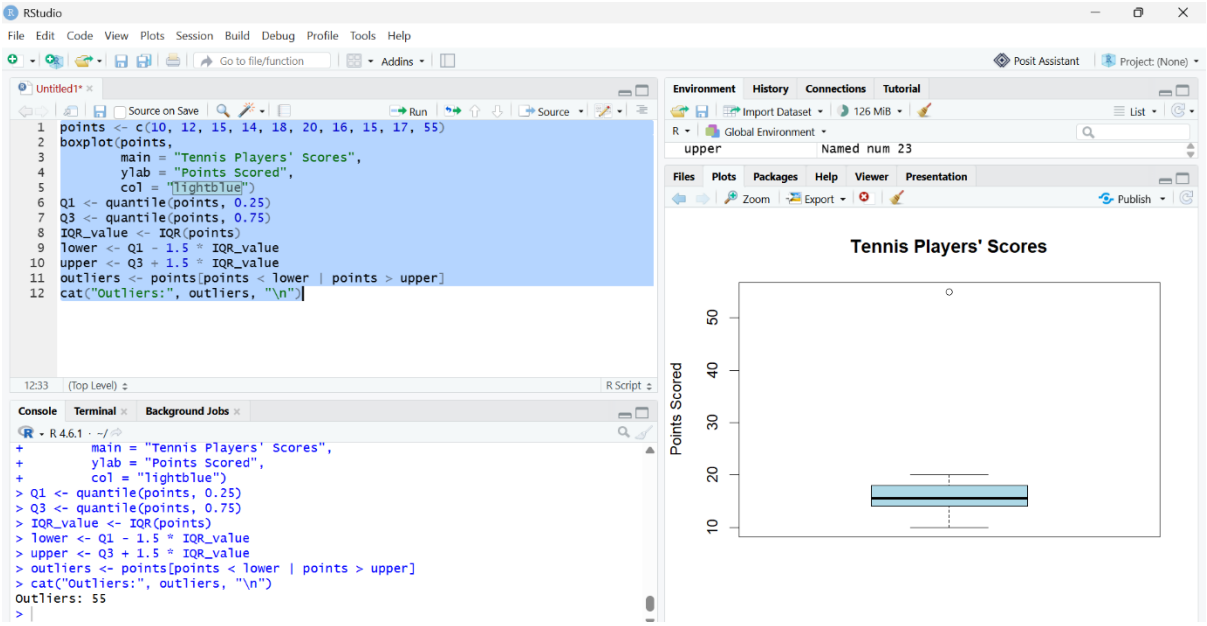
Q7.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
Source on Save
Run
Source
Breakpoints cannot be set until the file is saved.
12 plot(water$hardness, water$mortality,
13 main = "Hardness vs Mortality",
14 xlab = "Hardness",
15 ylab = "Mortality",
16 pch = 19,
17 col = "blue")
18 cat("Correlation:",
19 cor(water$hardness, water$mortality), "\n")
20 model <- lm(mortality ~ hardness, data = water)
21 print(summary(model))
22 abline(model, col = "red", lwd = 2)
23 prediction <- predict(
24 model,
25 newdata = data.frame(hardness = 88)
26)
27 cat("Predicted Mortality:", prediction, "\n")
27:46 (Top Level) R Script
Console Terminal Background Jobs
R 4.6.1 · ~/
A <- c(76, 35, 47, 64, 95, 66, 89, 36, 84)
A <- c(76, 35, 47, 64, 95, 66, 89, 36, 84)
Marks of both classes
Install MASS package (run only once if not installed)
install.packages("MASS")
library(MASS)
data("water", package = "MASS")
head(water)
Create sample water dataset
cat("Predicted Mortality:", prediction, "\n")
Environment History Connections Tutorial
Import Dataset 165 MiB
List
R Global Environment
mode "9"
num_bins 3
outliers 55
pencils num [1:10] 9 25 23 12 11 6 7 8 9 10
points num [1:10] 10 12 15 14 18 20 16 15 17 55
prices num [1:46] 1 1 5 5 5 5 5 8 8 10 ...
Q1 Named num 14.2
Q3 Named num 17.8
s 12.94
smoothed_data_by... Named num [1:48] 11 30 11 30 11 30 11 30 11 30...
smoothed_data_by... num [1:24(1d)] 18.9 18.9 18.9 18.9 18.9 ...
smoothed_data_by... num [1:24(1d)] 20 20 20 20 20 20 20 20 20 20 ...
upper Named num 23
v num [1:10] 78.3 81.8 82 74.2 83.4 84.5 82.9 77...
x num [1:10] 4 1 5 7 10 2 50 25 90 36
y num [1:10] 12 5 13 19 31 7 153 72 275 110
z_score num [1:5] -0.949 -0.632 -0.316 0.316 1.581
z_score_norm num [1:5] -0.949 -0.632 -0.316 0.316 1.581
Files Plots Packages Help Viewer Presentation
Zoom Export Publish

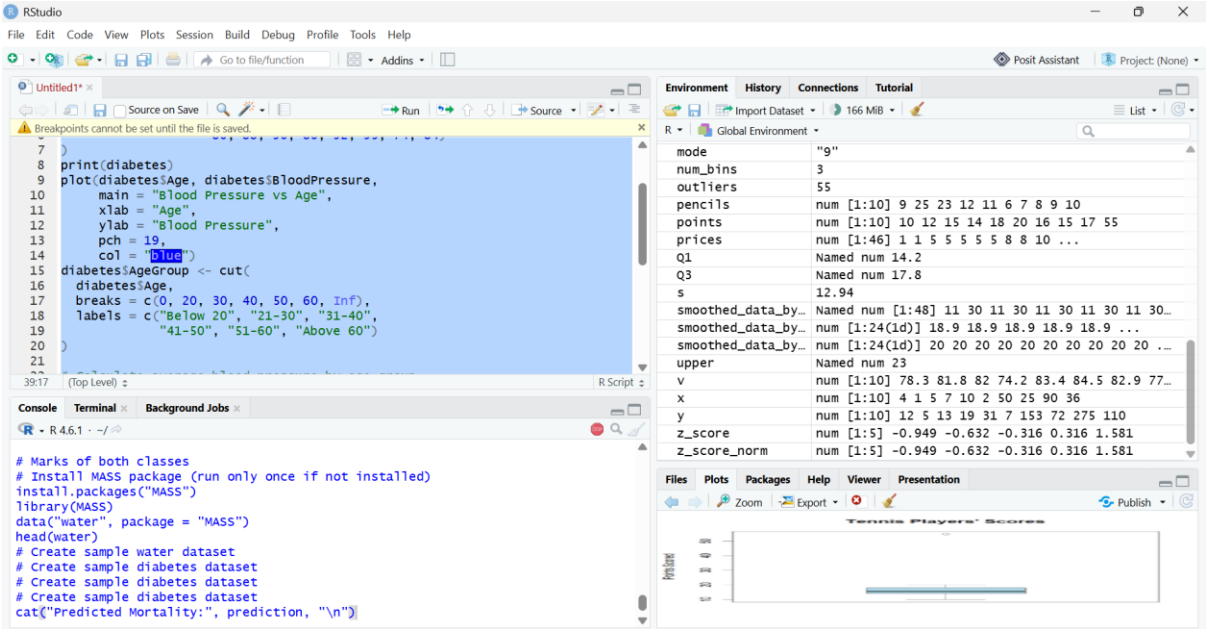
Q8.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
Source on Save
Run
Source
1 data(mtcars)
2 boxplot(mpg ~ factor(cyl),
3 data = mtcars,
4 main = "MPG by Number of Cylinders",
5 xlab = "Number of Cylinders",
6 ylab = "Miles Per Gallon (MPG)",
7 col = c("lightblue", "lightgreen", "orange"))
7:54 (Top Level) R Script
Console Terminal Background Jobs
R 4.6.1 · ~/
Error: object 'water' not found
Show Traceback
Rerun with Debug
> data(mtcars)
> boxplot(mpg ~ factor(cyl),
+ data = mtcars,
+ main = "MPG by Number of Cylinders",
+ xlab = "Number of Cylinders",
+ ylab = "Miles Per Gallon (MPG)",
+ col = c("lightblue", "lightgreen", "orange"))
>
Environment History Connections Tutorial
Import Dataset 127 MiB
List
R Global Environment
mtcars 32 obs. of 11 variables
Files Plots Packages Help Viewer Presentation
Zoom Export Publish
MPG by Number of Cylinders
Miles Per Gallon (MPG)
10
15
20
25
30
4
6
8
Number of Cylinders

Q9.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
points <- c(10, 12, 15, 14, 18, 20, 16, 15, 17, 55)
boxplot(points,
main = "Tennis Players' Scores",
ylab = "Points Scored",
col = "lightblue")
Q1 <- quantile(points, 0.25)
Q3 <- quantile(points, 0.75)
IQR_value <- IQR(points)
lower <- Q1 - 1.5 * IQR_value
upper <- Q3 + 1.5 * IQR_value
outliers <- points[points < lower | points > upper]
cat("Outliers:", outliers, "\n")
Console Terminal Background Jobs
+ main = "Tennis Players' Scores",
+ ylab = "Points Scored",
+ col = "lightblue")
> Q1 <- quantile(points, 0.25)
> Q3 <- quantile(points, 0.75)
> IQR_value <- IQR(points)
> lower <- Q1 - 1.5 * IQR_value
> upper <- Q3 + 1.5 * IQR_value
> outliers <- points[points < lower | points > upper]
> cat("Outliers:", outliers, "\n")
Outliers: 55
Environment History Connections Tutorial
upper Named num 23
Files Plots Packages Help Viewer Presentation
Tennis Players' Scores
Points Scored
10
20
30
40
50

Q10.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
Breakpoints cannot be set until the file is saved.
print(diabetes)
plot(diabetes$Age, diabetes$BloodPressure,
main = "Blood Pressure vs Age",
xlab = "Age",
ylab = "Blood Pressure",
pch = 19,
col = "blue")
diabetes$AgeGroup <- cut(
diabetes$Age,
breaks = c(0, 20, 30, 40, 50, 60, Inf),
labels = c("Below 20", "21-30", "31-40",
"41-50", "51-60", "Above 60")
)
Console Terminal Background Jobs
Marks of both classes
Install MASS package (run only once if not installed)
install.packages("MASS")
library(MASS)
data("water", package = "MASS")
head(water)
Create sample water dataset
Create sample diabetes dataset
Create sample diabetes dataset
Create sample diabetes dataset
cat("Predicted Mortality:", prediction, "\n")
Environment History Connections Tutorial
mode "9"
num_bins 3
outliers 55
pencils num [1:10] 9 25 23 12 11 6 7 8 9 10
points num [1:10] 10 12 15 14 18 20 16 15 17 55
prices num [1:46] 1 1 5 5 5 5 5 8 8 10 ...
Q1 Named num 14.2
Q3 Named num 17.8
s 12.94
smoothed_data_by... Named num [1:48] 11 30 11 30 11 30 11 30 11 30...
smoothed_data_by... num [1:24(1d)] 18.9 18.9 18.9 18.9 18.9 ...
smoothed_data_by... num [1:24(1d)] 20 20 20 20 20 20 20 20 20 20 ...
upper Named num 23
v num [1:10] 78.3 81.8 82 74.2 83.4 84.5 82.9 77...
x num [1:10] 4 1 5 7 10 2 50 25 90 36
y num [1:10] 12 5 13 19 31 7 153 72 275 110
z_score num [1:5] -0.949 -0.632 -0.316 0.316 1.581
z_score_norm num [1:5] -0.949 -0.632 -0.316 0.316 1.581
Files Plots Packages Help Viewer Presentation